

MIMIC BOT -A WIRELESS TELE-ROBOTIC SYSTEM VIA HAND LANDMARK TRACKING

THESIS

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SUBMITTED BY

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ABSTRACT

Tele-robotic systems play an important role in enabling remote operation and reducing human effort in complex and hazardous environments. However, traditional robotic systems often rely on manual controllers, which limit flexibility and natural interaction. To address these challenges, this project presents **MIMICBOT – A WIRELESS TELE-ROBOTIC SYSTEM VIA HAND LANDMARK TRACKING**, designed to achieve intuitive and real-time robotic control using human hand gestures. The system utilizes a camera-based vision module to capture hand movements and gesture patterns. These inputs are processed using computer vision techniques and transmitted wirelessly through an ESP32 microcontroller. The robotic arm replicates the human hand movements using servo motors, ensuring smooth, accurate, and synchronized motion. The system also incorporates an LM2596 DC-to-DC buck converter to provide a stable and regulated power supply to all components, improving system reliability and efficiency. The system operates in real time and provides a natural human–robot interaction experience. The implementation of this system demonstrates improved control accuracy, reduced dependency on physical controllers, and enhanced usability. Experimental results show that the robotic arm responds effectively to gesture inputs and performs precise movements. This project highlights the potential of integrating computer vision, embedded systems, and robotics to develop advanced tele-robotic solutions. Future enhancements may include the integration of artificial intelligence, improved gesture recognition accuracy, and applications in medical, industrial, and assistive robotics.

CHAPTER : 1

INTRODUCTION

1.1 INTRODUCTION TO TELE-ROBOTIC SYSTEMS

In recent years, robotics and automation have become increasingly important in various fields such as industry, healthcare, and research. Tele-robotic systems allow users to control machines remotely, making it possible to perform tasks in environments that are unsafe, inaccessible, or require high precision. Traditional robotic control methods usually depend on physical interfaces like joysticks or buttons, which can be complex to operate and do not provide a natural interaction experience. Therefore, there is a need for more intuitive and efficient control systems that can improve the way humans interact with robots.

Advancements in computer vision and wireless communication technologies have led to the development of gesture-based control systems. These systems enable robots to understand and respond to human hand movements without the need for wearable devices or direct physical contact. Hand landmark tracking techniques allow accurate detection of finger positions and gestures in real time, making the interaction more natural and user-friendly. The integration of such technologies with microcontrollers like ESP32 enables fast processing and wireless data transmission, allowing real-time control of robotic systems from a distance.

The project **MIMICBOT – A WIRELESS TELE-ROBOTIC SYSTEM VIA HAND LANDMARK TRACKING** focuses on developing a system that can replicate human hand movements using a robotic arm. The system captures hand gestures through a camera, processes the data using computer vision algorithms, and sends the information wirelessly to an ESP32 microcontroller. Based on the received data, the robotic arm moves accordingly using servo motors, achieving smooth and synchronized motion. A stable power supply is maintained using an LM2596 DC-to-DC buck converter to ensure reliable operation. This project demonstrates an effective combination of computer vision, embedded systems, and robotics to create an interactive and real-time tele-robotic solution with practical applications.

These limitations reduce efficiency and restrict the usability of robotic systems in real-world applications. Therefore, there is a need for an intelligent system that can provide natural, real-time, and wireless control of robotic devices using human gestures[2][5]

PROBLEM STATEMENT

Conventional robotic systems face several challenges due to their dependency on manual controllers and limited interaction capabilities. Some of the major problems include:

- Lack of natural and intuitive control methods
- Difficulty in operating robots in remote or hazardous environments
- Limited flexibility and responsiveness of traditional control systems
- High complexity and cost of advanced robotic systems
- Inability to accurately replicate human hand movements.

1.2 PROJECT SCOPE

The scope of this project includes the design and development of a wireless tele-robotic system capable of:

- Capturing human hand gestures using a camera-based system
- Processing gesture data using computer vision techniques
- Transmitting data wirelessly using ESP32
- Controlling a robotic arm to mimic human hand movements
- Providing real-time and synchronized robotic motion
- Ensuring stable power supply using LM2596 buck converter

This system can be applied in areas such as:

- Remote robotic operations
- Hazardous environment handling
- Assistive robotics
- Industrial training and automation
- Human–robot interaction research

1.3 OBJECTIVES OF THE PROJECT

The main objectives of this project are:

- To develop a wireless tele-robotic system using ESP32
- To implement hand landmark tracking for gesture recognition
- To control a robotic arm based on human hand movements
- To achieve real-time and accurate robotic motion
- To provide intuitive and user-friendly human–robot interaction
- To design a low-cost and efficient robotic system
- To ensure stable and reliable system operation using proper power management

1.4 OBJECTIVES OF THE PROJECT

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1.5 OVERVIEW OF MIMICBOT

MIMICBOT is a wireless tele-robotic system designed to control a robotic arm using human hand gestures. The system uses a camera-based vision module to capture hand movements and processes them using computer vision techniques. The processed data is transmitted wirelessly to an ESP32 microcontroller, which controls servo motors to replicate the hand movements in real time. A stable power supply is maintained using an LM2596 DC-to-DC buck converter, ensuring reliable system performance.

Advantages of the system:

- Provides natural and intuitive gesture-based control
- Enables real-time and wireless operation
- Improves accuracy and smoothness of robotic motion
- Reduces dependency on physical controllers
- Cost-effective and easy to implement
- Suitable for various applications like assistive robotics and remote operations

Thus, This project demonstrates the effective integration of modern technologies to create an intelligent and practical robotic system and highlights the potential of gesture-based control in improving human–machine interaction for real-world applications.

CHAPTER : 2

LITERATURE REVIEW

2.1 INTRODUCTION TO LITERATURE REVIEW

A literature review is an essential part of any project as it provides an understanding of the existing work carried out in a particular field. It helps in identifying the strengths and limitations of current systems and highlights the need for improvement. In the domain of tele-robotics and gesture-based control, various researchers have developed systems to enable intuitive human-machine interaction using different technologies such as sensors, computer vision, and wireless communication.

This chapter presents a study of existing gesture-controlled robotic systems, vision-based hand tracking techniques, and their limitations. It also emphasizes the need for an improved and integrated solution like the MimicBot system.[10][13]

2.2 EXISTING GESTURE-CONTROLLED SYSTEMS

Several gesture-controlled robotic systems have been developed to improve the interaction between humans and machines. Many of these systems use wearable devices such as data gloves, flex sensors, and accelerometers to capture hand movements. These sensors convert physical motion into electrical signals, which are then used to control robotic systems.

Some systems use microcontrollers like Arduino or Raspberry Pi to process sensor data and control robotic arms. Wireless communication technologies such as Bluetooth and Wi-Fi are often used to transmit data between the controller and the robot. These systems aim to provide remote control and automation capabilities.

Although wearable-based systems offer good accuracy, they have certain drawbacks such as discomfort for the user, high cost, and limited flexibility. Users are required to wear devices, which reduces convenience and restricts natural movement.

2.3 VISION-BASED HAND TRACKING SYSTEMS

With advancements in computer vision, camera-based hand tracking systems have gained popularity. These systems use image processing techniques to detect and track hand movements without the need for wearable devices. Technologies like OpenCV and MediaPipe enable real-time detection of hand landmarks, including finger positions and joint movements.

Vision-based systems provide a more natural and user-friendly interface, allowing users to control devices using simple hand gestures. These systems capture video input through a

camera, process the frames, and extract gesture information to control robotic systems.

However, these systems can be affected by environmental conditions such as lighting variations, background noise, and camera quality. In some cases, accuracy may decrease if the hand is partially occluded or moves too quickly.

2.4 LIMITATIONS OF EXISTING SYSTEMS

Despite the advancements in gesture-controlled and vision-based systems, several limitations still exist:

- Dependence on wearable devices in many systems
- Sensitivity to lighting and environmental conditions in vision-based systems
- Limited real-time performance and response delay
- High complexity and cost of implementation
- Lack of integration between different technologies
- Reduced accuracy due to noise and improper calibration

These limitations reduce the efficiency and reliability of existing systems and highlight the need for a better solution.

2.5 NEED FOR SYSTEM

The need for the system arises due to the limitations of existing systems. The MimicBot system aims to overcome these issues by providing an improved and efficient solution. The key needs are as follows:

- To eliminate the use of wearable devices and enable contactless gesture control
- To provide natural and intuitive human–robot interaction
- To achieve real-time and accurate control of robotic systems
- To reduce system complexity and improve ease of use
- To ensure reliable wireless communication using ESP32
- To improve system performance under different operating conditions
- To provide stable power supply using LM2596 DC-to-DC buck converter
- To enhance flexibility for applications such as assistive robotics and remote operation

CHAPTER : 3

EXISTING SYSTEM & METHODOLOGY

3.1 EXISTING SYSTEM DESCRIPTION

Existing robotic control systems mainly rely on traditional input devices such as joysticks, buttons, or wired controllers. These systems require direct human involvement and often lack flexibility in operation. In some advanced systems, wearable devices such as sensor gloves are used to capture hand movements and control robotic arms. These gloves use sensors like flex sensors and accelerometers to detect finger movements and transmit signals to the robotic system.[8]

In recent developments, vision-based systems have been introduced, where cameras are used to capture hand gestures and control robots using image processing techniques. These systems use computer vision algorithms to detect hand movements and convert them into control signals. While these systems provide a more natural interaction, they still face challenges in terms of accuracy, real-time performance, and environmental dependency.

Overall, existing systems provide basic functionality but lack integration, efficiency, and ease of use required for advanced real-time applications.[9]

3.2 LIMITATIONS OF EXISTING SYSTEM

The existing systems suffer from several drawbacks, including:

- **Limited Interaction:** Most systems rely on joysticks or wearable devices, which reduce natural human–robot interaction.
- **Lack of Flexibility:** Traditional control methods do not provide smooth and intuitive control of robotic movements.
- **Environmental Dependency:** Vision-based systems are affected by lighting conditions and background disturbances.
- **High Cost:** Advanced robotic systems with wearable devices and sensors increase overall system cost.
- **Delay in Response:** Some systems do not provide real-time performance, leading to slower response.

These limitations reduce the efficiency, usability, and reliability of existing robotic systems.

3.3 NEED FOR SYSTEM

Due to the drawbacks of existing systems, there is a need for an improved solution that can:

- Provide contactless and intuitive gesture-based control
- Eliminate the use of wearable devices
- Ensure real-time and accurate robotic movement
- Reduce system complexity and cost
- Improve user interaction and flexibility
- Enable wireless communication for remote operation
- Ensure stable and reliable system performance

The MimicBot system is designed to address these requirements effectively.

3.4 METHODOLOGY

The system, **MIMICBOT**, is a wireless tele-robotic system that uses hand landmark tracking for controlling a robotic arm in real time. The system integrates computer vision, embedded systems, and wireless communication to achieve efficient and accurate control.

Working Methodology:

- **Data Acquisition:**

A camera captures real-time video of hand movements.

- **Gesture Processing:**

Hand landmarks are detected using computer vision techniques such as MediaPipe, and gesture data is extracted.

- **Data Transmission:**

The processed data is transmitted wirelessly using ESP32.

- **Control Mechanism:**

The ESP32 microcontroller processes the received data and generates control signals.

- **Robotic Movement:**

Servo motors move the robotic arm according to the detected hand gestures.

- **Power Management:**

An LM2596 DC-to-DC buck converter provides stable voltage to ensure reliable operation of all components.

- **Continuous Operation:**

The system continuously captures, processes, and controls the robotic arm in real time.

3.5 ADVANTAGES OF METHODOLOGY

1. Provides real-time gesture-based control
2. Eliminates the need for wearable devices
3. Improves accuracy and smoothness of motion
4. Enables wireless and remote operation
5. Easy to use and user-friendly system
6. Cost-effective and scalable design
7. Reliable performance with stable power supply

CHAPTER : 4

SYSTEM DESIGN

4.1 Overview of System

The system is a Gesture-Controlled Robotic Hand System (Mimic Bot) designed to replicate human hand movements in real time using computer vision and embedded technology. The system integrates image processing techniques, a microcontroller, and actuator mechanisms to achieve intuitive human-machine interaction.

The system uses a laptop camera to capture hand gestures, which are processed using advanced computer vision libraries such as OpenCV and MediaPipe. These tools detect and track hand landmarks, enabling the identification of finger positions and movements.

The ESP32 Dev Board acts as the central control unit, receiving processed gesture data from the laptop via serial communication. Based on the received data, the ESP32 generates PWM signals to control multiple SG90 Servo Motor units, which are responsible for the movement of the robotic fingers and wrist.

The system employs a tendon-driven mechanism where servo motors pull nylon threads to bend the fingers, while rubber bands restore them to their original position. A regulated power supply is provided using an LM2596 to ensure stable operation of the servo motors.

The system enables real-time, contactless control of a robotic hand, making it suitable for applications in robotics, assistive technology, and automation.[1][20]

4.2 System Architecture

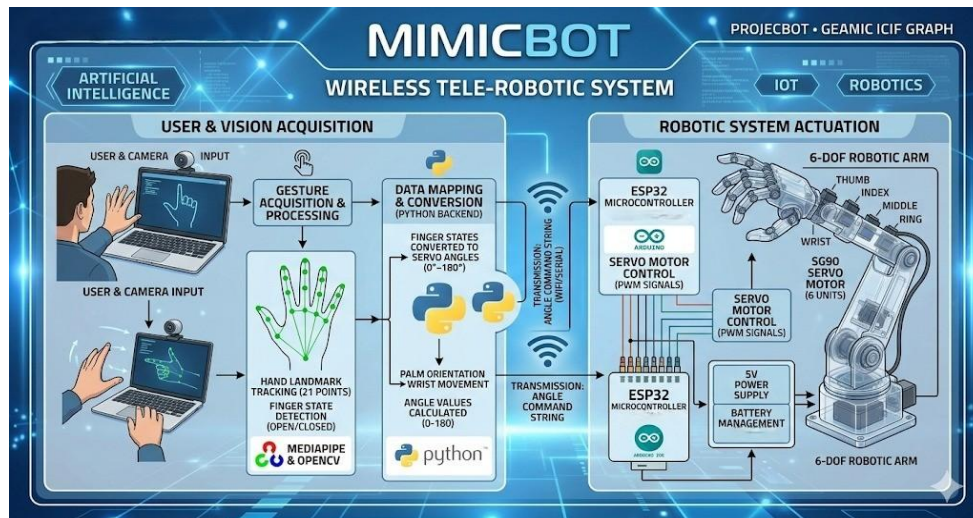


Fig 4.1 : System Architecture

The system architecture consists of three main components:

1. Input Layer (Gesture Input):

- Laptop Camera (Real-time video capture)
 - MediaPipe (Hand tracking and landmark detection)
 - OpenCV (Image processing)
 - These components continuously capture and process human hand gestures.
-

2. Processing Layer:

- ESP32 Dev Board
 - Python Processing Unit (Gesture analysis)
 - The Python program processes hand landmarks and generates command strings.
 - The ESP32 receives this data and converts it into control signals for actuators.
-

3. Output Layer (Actuators):

- SG90 Servo Motor (Finger Movement)
- Wrist Servo Motor (Hand Orientation)
- Robotic Hand Mechanism (Cardboard structure with tendon system)
- Servo motors move according to gesture input to mimic human hand movement.

4.3 Block Diagram

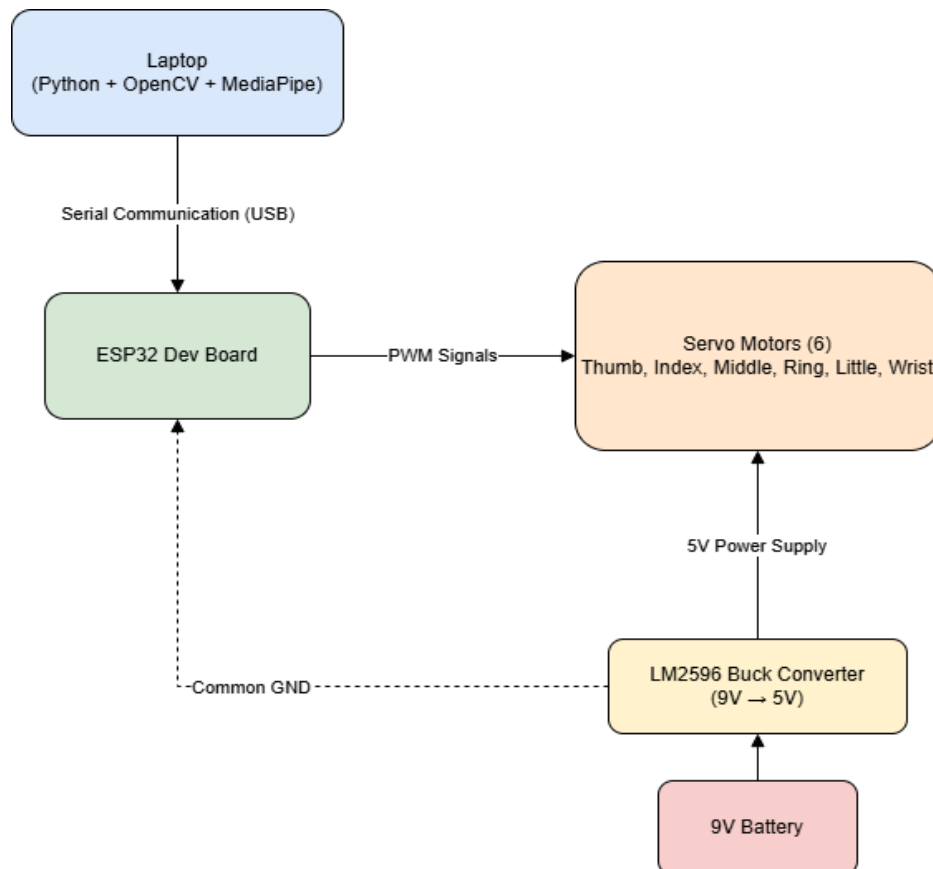


Fig 4.2 : Block Diagram

4.4 Working Principle

The working of the system is based on real-time gesture detection and robotic control.

- The system starts when power is supplied to the ESP32 and the Python program is executed on the laptop.
- The camera is initialized and begins capturing live video frames.
- Hand detection is performed using MediaPipe.
- Hand landmarks (21 points) are extracted from each frame.
- The landmark data is processed using OpenCV to determine finger positions.
- The system identifies whether each finger is open or closed.
- Based on finger states, corresponding servo motor angles are calculated.
- Palm orientation is analyzed to determine wrist movement.
- A command string is generated:

T: , I: , M: , R: , L: , W:

- The command string is transmitted to the ESP32 Dev Board via serial communication.
- ESP32 receives and processes the command data.
- PWM signals are generated to control multiple SG90 Servo Motor units.
- Based on conditions:

If finger is open → Corresponding servo rotates to open position

If finger is closed → Corresponding servo rotates to closed position

If hand tilts left/right → Wrist servo adjusts position

- The servo motors actuate the robotic hand using a tendon mechanism.
- The system continuously repeats this process in real time.

CHAPTER : 5

HARDWARE COMPONENTS

5.1 INTRODUCTION

The system consists of various hardware components including microcontrollers, sensors, actuators, and power management units. These components work together to capture hand gestures, process data, and control the robotic arm. Each component plays an important role in ensuring accurate and reliable operation of the system.[13][19]

5.2 ESP32 MICROCONTROLLER

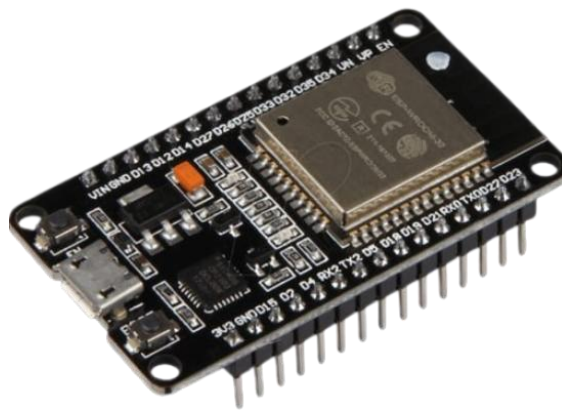


Fig 5.1 : ESP32 Board

- The ESP32 is the main controller used in the system.
- It has built-in Wi-Fi and Bluetooth capabilities
- Processes data received from sensors
- Sends control signals to actuators via relay module
- Enables communication with mobile applications
- Features:

Dual-core processor Low
power consumption High
processing speed
Multiple GPIO pins

5.3 SERVO MOTORS



Fig 5.12 : Servo motor

- The servo motor controls ventilation windows.
- Provides precise angular movement
- Opens/closes vents
- Controlled by ESP32

5.4 LM2596 BUCK CONVERTER

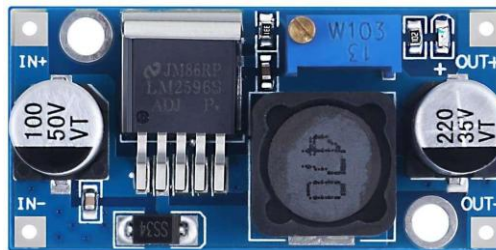


Fig 5.14 : LM2596 Buck Converter

- The LM2596 is used to step down voltage from 9V to 5V.
- Provides stable power supply to all components.
- Improves efficiency and reduces heat loss.
- Adjustable output voltage using onboard potentiometer.
- Ensures safe operation of ESP32, sensors, and relay module.

5.5 BATTERY (9V POWER SUPPLY)



Fig 5.14 : Battery

- The 9V battery is used as the main power source for the system.
- It provides input voltage to the LM2596 buck converter.
- Ensures portability and easy power supply for the project.
- Suitable for low-power robotic and embedded systems.

5.8 SUMMARY OF HARDWARE COMPONENTS

All the All the above components are integrated to form the complete MimicBot system. The camera captures hand gestures, the ESP32 processes the data, and the servo motors control the robotic arm accordingly. The LM2596 ensures a stable power supply, enabling efficient and reliable system performance.

CHAPTER :6

SOFTWARE & IMPLEMENTATION

6.1 INTRODUCTION

This chapter describes the software tools, programming techniques, and implementation process used in the development of the **MIMICBOT** system. The system integrates computer vision, data processing, and wireless communication to enable real-time gesture-based control of a robotic arm. The software plays a crucial role in capturing hand gestures, processing data, and transmitting commands to the robotic system.[14][16][15]

6.2 SOFTWARE TOOLS USED

The following software tools are used for implementation:

- **Python Programming:**
Used for gesture processing, data mapping, and communication.
- **OpenCV Library:**
Used for image processing and capturing video input from the camera.
- **MediaPipe:**
Used for hand landmark detection and tracking (24 key points).
- **Arduino IDE:**
Used for programming the ESP32 microcontroller.
- **Wireless Communication (Wi-Fi):**
Used for transmitting data between the computer and ESP32.

6.3 GESTURE ACQUISITION & PROCESSING

The system captures real-time hand gestures using a camera connected to a computer.

The video frames are processed using OpenCV and MediaPipe to detect hand landmarks.

- The camera captures live video input.
- MediaPipe detects 24 hand landmark points.
- Finger states (open/close) are identified.
- Hand movements are tracked continuously.

This process enables accurate detection of hand gestures in real time.

6.4 DATA MAPPING & CONVERSION

After detecting hand landmarks, the system processes the data to convert it into usable control signals.

- Finger positions are converted into servo angles (0° – 180°).
- Palm orientation and wrist angles are calculated.
- Gesture data is processed using Python.
- A command string containing angle values is generated.

This step ensures proper mapping between human hand movement and robotic motion.

6.5 WIRELESS COMMUNICATION

The processed data is transmitted wirelessly to the ESP32 microcontroller.

- Wi-Fi is used for communication.
- Angle command string is sent from the computer to ESP32.
- Real-time data transfer ensures minimal delay.
- Enables remote operation of the robotic arm

6.6 ROBOTIC CONTROL SYSTEM

The ESP32 receives the transmitted data and controls the robotic arm accordingly.

- ESP32 processes received angle values.
- PWM signals are generated for servo motors.
- Each servo motor moves according to the assigned angle.
- Robotic arm mimics human hand movements.

6.7 POWER MANAGEMENT

The ESP32 receives the transmitted data and controls the robotic arm accordingly.

- A 9V battery provides input power..
- LM2596 buck converter steps down voltage to 5V.
- Provides regulated supply to ESP32 and servo motors.
- Ensures system stability and prevents damage.

6.8 SYSTEM INTEGRATION

The complete system is divided into two main parts:

1. User & Vision Acquisition System:

Captures and processes hand gestures using camera and software.

2. Robotic System Actuation:

Receives commands and controls the robotic arm using ESP32.

Both systems work together to provide real-time gesture-based control.

6.9 ADVANTAGES OF IMPLEMENTATION

- Real-time gesture-based control
- High accuracy using hand landmark tracking
- Wireless and flexible operation
- User-friendly interface
- Efficient and low-cost system

6.10 SUMMARY

The software implementation of MimicBot successfully integrates computer vision, data processing, and wireless communication to control a robotic arm in real time. The system provides accurate gesture recognition and smooth robotic movement, demonstrating an effective and practical tele-robotic solution.

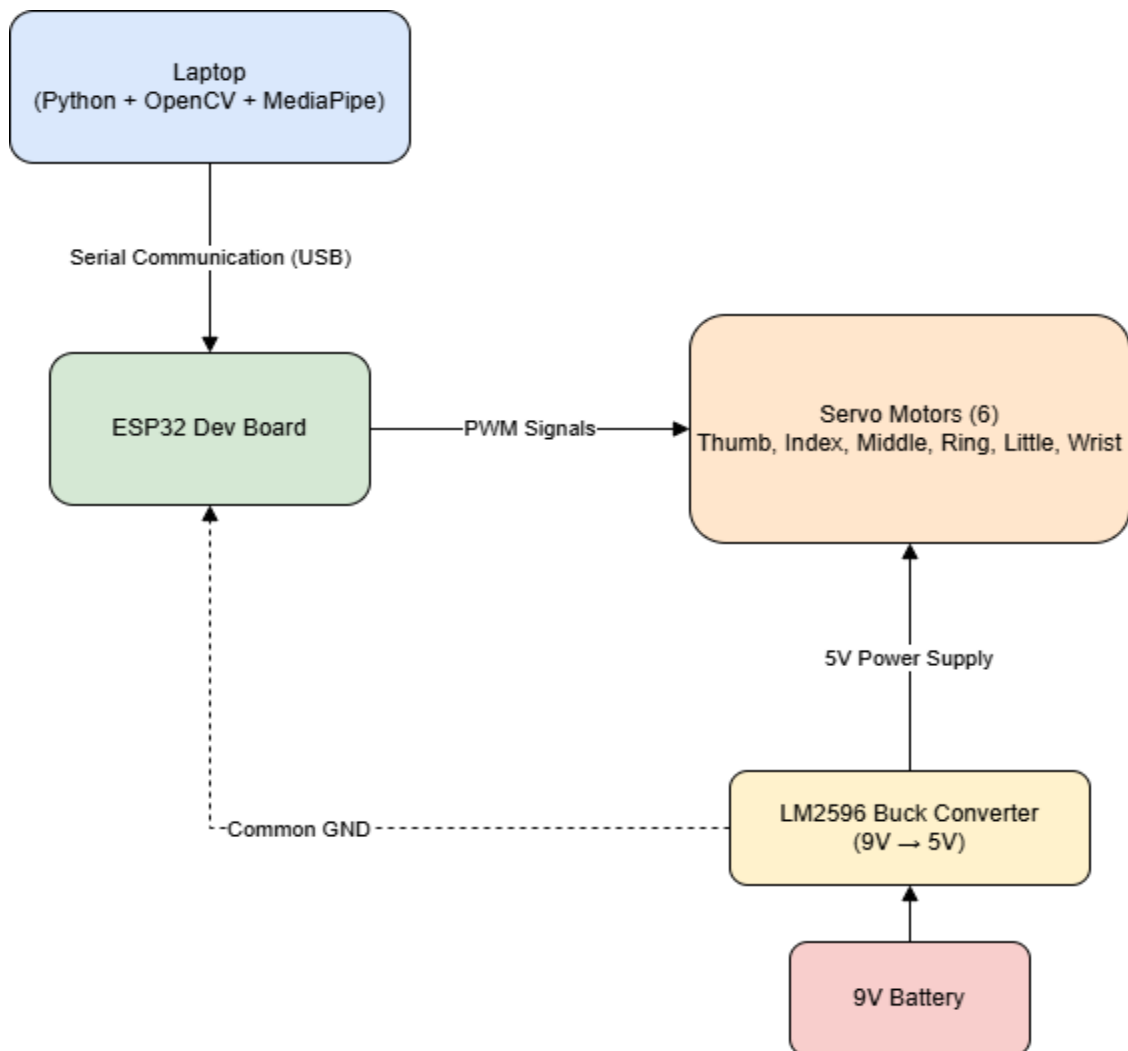
CHAPTER : 7

CIRCUIT DIAGRAM & WORKING

7.1 Introduction

This chapter explains the circuit connections and working of the Gesture-Controlled Robotic Hand System (Mimic Bot). The system integrates computer vision techniques with the ESP32 Dev Board microcontroller to achieve real-time gesture-based control of a robotic hand. Hand gestures are captured using a camera and processed through software using OpenCV and MediaPipe. The processed data is converted into servo motor control signals and transmitted to the ESP32 via serial communication. The ESP32 generates PWM signals to control multiple SG90 Servo Motor units, which actuate the movement of the robotic fingers and wrist. A regulated power supply is provided using an LM2596 to ensure stable operation of the system. This integration enables accurate, real-time replication of human hand movements through the robotic arm.[18][4]

7.2 Circuit Diagram



7.3 Circuit Connections

- The following connections are used in the Mimic Bot system:

1. ESP32 Connections

- Acts as the main controller of the system.
- Receives gesture data from laptop and controls servo motors.
- All servo motors are connected to its GPIO pins.

Component	GPIO Pin
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Thumb Servo	13
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Index Servo	14
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Middle Servo	25
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Ring Servo	26
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Little Servo	27
--------------	----

Wrist Servo	33
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2. Servo Motor Connections

- Six SG90 Servo Motor are used for finger and wrist movement.

Each servo has three wires:

- Red → VCC (5V)
- Brown → GND
- Yellow → Signal (GPIO pin)

Connections:

- Signal wires → Connected to respective ESP32 GPIO pins
- VCC → 5V regulated supply (from buck converter)
- GND → Common ground

3. Serial Communication Connection

- Communication between laptop and ESP32 is done using USB cable.
 - Laptop (Python Program) → USB → ESP32 Dev Board
 - Baud Rate: 115200

4. Power Supply

- Main Supply:

9V Battery → LM2596 → 5V regulated output

- Distribution:

- Servo motors powered from 5V output
- ESP32 powered via USB
- GND of ESP32 and power supply must be common

- Important:

All GND connections must be common to ensure proper operation of the system.

7.4 Step-by-Step Working

The system works in the following sequence:

1. Power ON System
2. ESP32 and Python program initialize
3. Camera Activation
4. Laptop camera captures real-time video
5. Hand Detection
6. Hand is detected using MediaPipe
7. Landmark Extraction
8. 21 hand landmark points are identified
9. Finger State Detection
10. Each finger is identified as open or closed
11. Angle Calculation
12. Finger states are converted into servo angles
13. Wrist Movement Detection
14. Palm orientation is used to calculate wrist angle
15. Command Generation
16. Command string is created:
 T:90 , I:0 , M:90 , R:0 , L:0 , W:135
17. Data Transmission
18. Command is sent to ESP32 via serial communication
19. Data Processing (ESP32)
20. ESP32 parses received command string
21. PWM Signal Generation
22. ESP32 generates PWM signals
23. Servo Movement
24. Servo motors rotate accordingly
25. Robotic Hand Movement
26. Fingers and wrist mimic human hand
27. Continuous Loop
28. System repeats continuously in real time

7.5 Working Modes

1. Real-Time Mode

- System works continuously based on live hand gestures
- No delay between input and output

2. Testing Mode

- Manual commands can be sent via Serial Monitor
- Used for debugging servo movement

7.6 Safety and Reliability

- Buck converter ensures stable voltage supply
- Common ground prevents signal errors

- Controlled data transmission avoids system freezing
- Proper wiring ensures safe operation of ESP32 and servos

7.7 Summary

The circuit design of the Mimic Bot ensures proper communication between the software and hardware components. The system efficiently converts human hand gestures into servo motor movements using ESP32 and real-time processing. The use of regulated power supply and structured communication enables stable and reliable operation of the robotic hand.

CHAPTER : 8

RESULTS & ANALYSIS

8.1 INTRODUCTION

This chapter presents the results obtained from the implementation of the **MIMICBOT** system. It includes the performance of gesture recognition, robotic movement, and overall system functionality. The analysis is based on real-time testing of the system under different conditions.

8.2 SYSTEM OUTPUT

The system successfully captures and processes hand gestures and controls the robotic arm accordingly.

Observed Outputs:

- Real-time detection of hand landmarks
- Accurate tracking of finger movements
- Smooth and synchronized movement of robotic arm
- Wireless transmission of data without significant delay
- Proper response of servo motors based on gesture input

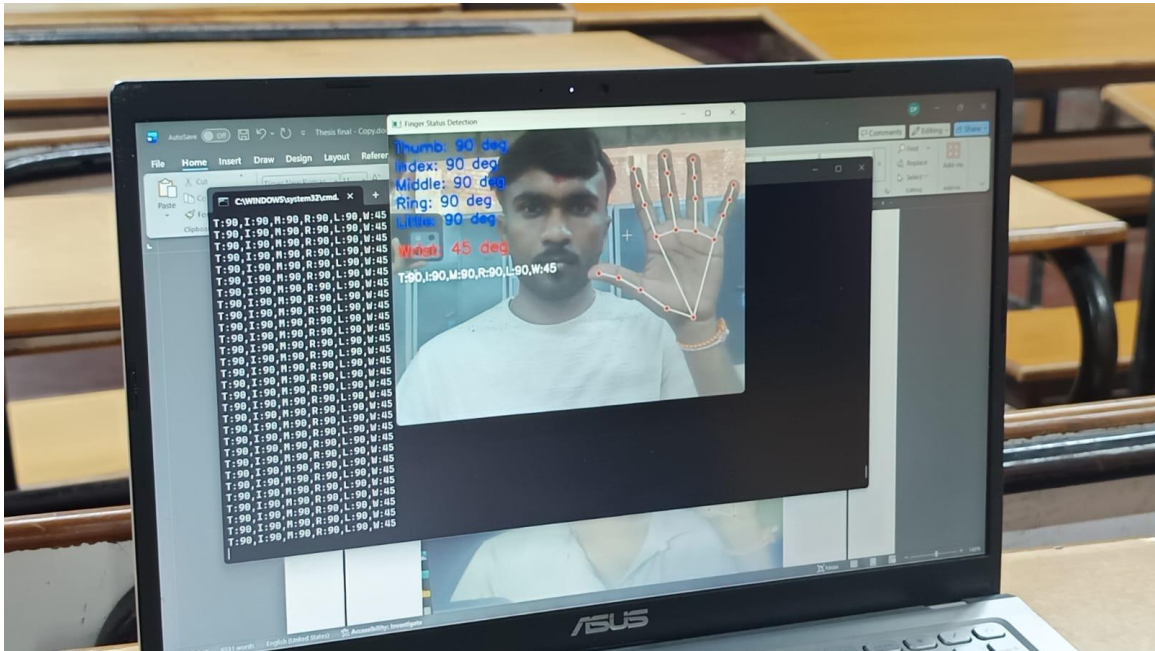


Fig.8.1: Hand motion tracking

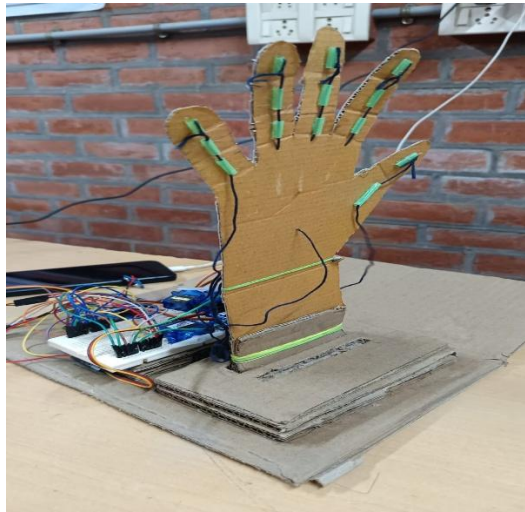


Fig.8.2: Prototype Model

```
T:0,I:0,M:0,R:0,L:0,W:90
T:0,I:0,M:0,R:0,L:0,W:90
T:0,I:0,M:0,R:0,L:0,W:90
T:0,I:0,M:0,R:0,L:0,W:90
T:0,I:0,M:0,R:0,L:0,W:90
T:0,I:90,M:0,R:0,L:0,W:90
T:0,I:0,M:0,R:0,L:0,W:90
T:0,I:0,M:0,R:0,L:0,W:90
T:0,I:0,M:0,R:0,L:0,W:135
T:0,I:90,M:0,R:0,L:0,W:90
```

Fig8.3: Serial output of the command

8.3 PERFORMANCE ANALYSIS

The performance of the system is evaluated based on the following parameters:

- **Accuracy:**
The system accurately detects hand gestures using MediaPipe and converts them into precise robotic movements.
- **Response Time:**
The system shows fast response due to efficient processing and wireless communication using ESP32.
- **Smoothness of Motion:**
Servo motors provide smooth and continuous movement of the robotic arm.
- **Reliability:**
The system performs consistently under normal lighting conditions and stable network connectivity.

- **Efficiency:**
The system reduces manual effort and provides an intuitive control mechanism

8.4 ADVANTAGES OF THE SYSTEM

- Real-time gesture-based robotic control
- Natural and intuitive user interaction
- Wireless and flexible operation
- High accuracy using computer vision
- Cost-effective and easy to implement

8.5 LIMITATIONS

- Performance depends on lighting conditions
- Requires stable Wi-Fi connection
- Accuracy may decrease with fast hand movement
- Limited range based on network coverage
- Initial calibration is required for better performance

8.6 SUMMARY

The results demonstrate that the MimicBot system works effectively in real-time gesture recognition and robotic control. The system achieves good accuracy, fast response, and smooth operation, making it suitable for practical applications. However, certain limitations such as environmental dependency and network constraints need to be addressed for further improvement.

CHAPTER : 9

CONCLUSION & FUTURE SCOPE

9.1 CONCLUSION

The **MIMICBOT – A WIRELESS TELE-ROBOTIC SYSTEM VIA HAND LANDMARK TRACKING** has been successfully designed and implemented. The system integrates computer vision, embedded systems, and wireless communication to enable gesture-based control of a robotic arm in real time. It provides an intuitive method of interaction without the need for physical controllers.

The system uses hand landmark tracking to accurately detect finger movements and convert them into control signals. These signals are transmitted wirelessly to the ESP32 microcontroller, which controls the servo motors of the robotic arm. The implementation demonstrates improved performance in terms of accuracy, response time, and ease of use. The use of an LM2596 DC-to-DC buck converter ensures stable power supply, enhancing system reliability. Overall, the project provides a cost-effective and efficient solution for real-time tele-robotic control.

This project highlights the potential of integrating modern technologies to develop intelligent robotic systems. It also shows how gesture-based control can improve human–robot interaction and make robotic systems more accessible and user-friendly.

9.2 FUTURE SCOPE

Although the system performs efficiently, there are several enhancements that can be implemented in the future:

- Use of machine learning algorithms for adaptive and predictive control
- Implementation of cloud connectivity for data storage and remote monitoring
- Development of a mobile application for improved user interface and control
- Expansion to support robotic arms with higher degrees of freedom (DOF)
- Improvement in system performance under varying lighting conditions
- Application in medical fields such as robotic surgery and rehabilitation
- Use in industrial automation for complex and precise operations

These improvements can make the system more intelligent, scalable, and efficient for future agricultural applications.

CHAPTER : 10

APPLICATIONS

10.1 INTRODUCTION

The MIMICBOT – A WIRELESS TELE-ROBOTIC SYSTEM VIA HAND LANDMARK TRACKING has a wide range of applications in various fields due to its ability to provide real-time, intuitive, and wireless control of robotic systems. The integration of computer vision and gesture-based control makes it suitable for both industrial and research-based applications. This chapter describes the major application areas where the system can be effectively utilized.

10.2 APPLICATIONS OF THE SYSTEM

1. REMOTE ROBOTIC OPERATIONS

The system can be used to control robots remotely in environments that are unsafe or difficult for humans to access. For example, it can be applied in disaster management, nuclear plants, or deep-sea operations. The gesture-based control allows operators to perform tasks naturally without complex controllers.

2. MEDICAL AND SURGICAL APPLICATIONS

MimicBot can be used in medical fields such as robotic surgery and rehabilitation systems. Doctors can control robotic arms remotely using hand gestures for precise operations. It can also be used in physiotherapy devices to assist patients in performing guided movements.

3. INDUSTRIAL AUTOMATION

In industries, the system can be used for pick-and-place operations, assembly lines, and handling hazardous materials. It reduces human effort and increases efficiency by allowing operators to control robotic arms intuitively. This improves productivity and workplace safety.

4. ASSISTIVE ROBOTICS

The system can help physically challenged individuals by enabling them to control robotic arms for daily tasks such as picking objects or interacting with their surroundings. It provides independence and improves quality of life through easy and natural control mechanisms.

5. EDUCATION AND RESEARCH

MimicBot can be used as a learning tool in robotics, IoT, and computer vision fields. Students and researchers can use it to understand gesture recognition, embedded systems, and robotic control. It is useful for academic projects and experimental research.

6. HUMAN-ROBOT INTERACTION (HRI)

The system enhances human-robot interaction by providing a natural way of communication through gestures. It can be used in research related to AI and robotics to develop more advanced interactive systems.

7. DEFENSE AND SURVEILLANCE

In defense applications, the system can be used to control robotic arms for bomb disposal, surveillance, and handling dangerous objects. It minimizes human risk and improves

operational safety.

8. ENTERTAINMENT AND GAMING

The gesture-based control system can be used in gaming and virtual environments to control characters or robotic devices. It provides an interactive and immersive experience for users.

10.3 SUMMARY

The MimicBot system has diverse applications across multiple domains including healthcare, industry, education, and defense. Its ability to provide real-time, wireless, and intuitive control makes it a versatile and practical solution for modern technological challenges.

CHAPTER : 11

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